

Agenda

1. Friday Feedback
2. Integration By Parts
(continued)



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Recall: $\int f(x)g'(x)dx = f(x) \cdot g(x) - \int g(x) \cdot f'(x)dx$

$$\int u dv = uv - \int v du$$

Activity: $\int x \cos x dx$

$$\begin{aligned} dv &= \cos(x) & v &= \sin(x) \\ u &= x & du &= dx \end{aligned}$$

$$\begin{aligned} &= x \sin(x) - \int \sin(x) dx \\ &= \boxed{x \sin(x) + \cos(x) + C} \end{aligned}$$

Integration by Parts: Guidelines for choosing u and dv

- dx is always part of ~~du~~ dv
- dv should be easy to integrate
- u and dv are chosen so that $\int v du$ is no more difficult to integrate than the original integral
- If the new integral is more complicated, try different choices for u and dv

Example: $\int \arctan(x) dx$

$$dv = dx \quad v = x$$

$$u = \arctan x \quad du = \frac{1}{1+x^2} dx$$

$$= x \tan^{-1}(x) - \int \frac{x}{1+x^2} dx \quad \begin{array}{l} u = 1+x^2 \\ du = 2x \end{array}$$

$$= x \tan^{-1}(x) - \frac{1}{2} \ln |u| + C$$

$$= x \tan^{-1}(x) - \frac{1}{2} \ln(1+x^2) + C$$

A general rule of thumb is that when

choosing u , the order of priority follows

the acronym

LIATE

Logarithm
Inverse trig
Algebraic
Trig
Exponential

L > make really bad
I > du 's
hard to integrate
directly

A > degree goes down
so nice to differentiate

T > differentiation doesn't
make things simpler

Activity: $\int x^2 \sin x dx$

$$u = x^2 \quad du = 2x dx$$

$$dv = \sin(x) dx \quad v = -\cos(x)$$

$$= -x^2 \cos(x) + 2 \int \cos(x) x dx$$

$$u = x \quad du = dx$$

$$dv = \cos(x) \quad v = \sin(x)$$

$$= -x^2 \cos(x) + x \sin(x) + 2 \int \sin(x) dx$$

$$= -x^2 \cos(x) + 2x \sin(x) + 2 \cos(x) + C$$

$$= -x^2 \cos(x) + 2(x \sin x + \cos x) + C$$

Example: $\int e^{2x} \cos(3x) dx$

$$\begin{array}{l} u = e^{2x} \\ dv = \cos(3x) dx \\ du = 2e^{2x} dx \\ v = \frac{1}{3} \sin(3x) \end{array} \left\{ \begin{array}{l} = \frac{1}{3} e^{2x} \sin(3x) \\ - \frac{2}{3} \int e^{2x} \sin(3x) \end{array} \right.$$

$$\frac{1}{3} e^{2x} \sin(3x) - \frac{2}{3} \int e^{2x} \sin(3x)$$

$$\frac{1}{3} e^{2x} \sin(3x) + \frac{2}{3} \left[-\frac{1}{3} e^{2x} \cos(3x) - \frac{2}{3} \int e^{2x} \cos(3x) dx \right]$$

$$\int e^{2x} \cos(3x) dx = \frac{1}{3} e^{2x} \sin(3x) + \frac{2}{9} e^{2x} \cos(3x) - \frac{4}{9} \int e^{2x} \cos(3x)$$

$$\frac{9}{9} \int e^{2x} \cos(3x) dx + \frac{4}{9} \int e^{2x} \cos(3x) dx = \frac{1}{3} e^{2x} \sin(3x) + \frac{2}{9} e^{2x} \cos(3x) + \dots$$

$$\frac{9}{13} \left(\frac{13}{9} \int e^{2x} \cos(3x) dx \right) = \left(\frac{1}{3} e^{2x} \sin(3x) + \frac{2}{9} e^{2x} \cos(3x) \right) \frac{9}{13}$$

$$\int e^x \cos(3x) dx = \frac{1}{13} \left[\frac{1}{3} e^{2x} \sin(3x) + \frac{2}{9} e^{2x} \cos(3x) \right] + C$$

Find a definite integral using Integration by Parts

Method 1:

Method 2:

Example: $\int_0^{\pi/4} x \sec^2 x dx$

Activity $\int_1^e (\ln x)^2 dx$
(if time)

Reduction Formulas

Example: Prove the reduction formula

$$\int x^n \arcsin x \, dx = \frac{x^{n+1}}{n+1} \arcsin x - \frac{1}{n+1} \int \frac{x^{n+1}}{\sqrt{1-x^2}} \, dx$$

Activity: $\int x\sqrt{x+1} dx$
(if time)